

Characterising macropores and preferential flow of mountainous forest soils with contrasting human disturbances

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Abstract. Preferential flow can develop in soil macropores, and macropores are sensitive to human disturbances. This study investigated soil macropore features and the main factors controlling preferential flow at four sites with different levels of human disturbance in a mountainous area in Central China. The level of human disturbance decreased with increasing elevation, with the lowest elevation areas covered with coniferous trees (LF) > middle mountain areas covered with tea gardens (TG) > middle mountain areas covered with deciduous trees and mixed shrubs (MF) > subalpine areas covered with evergreen coniferous trees (HF). At each site, the soil macropore structure at 0–20 cm soil depth was analysed using computed tomography scans (0.6 mm resolution) and Image J software. Preferential flow was determined by analysing the breakthrough curve (BTC) of nitrate. The macroporosity, surface area density, mean macropore size, macropore number density, length density and node density were all ranked in the order of HF ≥ MF ≥ TG = LF. Less disturbed sites had stronger evidence of preferential flow as shown by faster breakthrough, longer tails and greater asymmetry of the BTCs. There were significant ($P < 0.05$) positive influences of soil macropore properties on pore water velocity and the solute dispersion coefficient. The dispersivity parameter was mainly affected by the macropore equivalent hydraulic radius. This study showed that human disturbance in the mountain forest areas significantly decreased soil macropores by changing soil physical properties (e.g. bulk density, texture and soil organic matter content) and root distribution, thus increasing the risk of surface runoff and nutrient losses.

Additional keywords: breakthrough curve, CDE equation, computed tomography, MIM model, root distribution, soil physical properties.

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Introduction

Soil macropores play a crucial role in soil water infiltration and solute transport (Beven and Germann 1982; Luo *et al.* 2008; Muller *et al.* 2018). Preferential flow can occur through macropores, enabling rainwater to migrate faster from soil surface to subsurface layers and to underground water sources (Luxmoore *et al.* 1990). Macropores can be developed from decayed roots, earthworm activity and soil swelling or shrinking cycles in a natural ecosystem. Macropores can be destroyed by human activities such as cultivation or deforestation (Liu *et al.* 2018). Although macroporosity usually accounts for a small portion of total porosity, macropores can transport a large proportion of infiltrated soil water (Li *et al.* 2009). Macropores and related preferential flow can transport rainwater to subsurface soil, reducing the occurrence of surface runoff and soil erosion along steep slopes, which helps to prevent geological hazards, such as landslides or debris in mountainous areas.

The Three Gorges mountain area is located at the transitional zone of the middle-lower Yangtze Plains and the Tibetan Plateau. The topography mostly consists of gullies, valleys, hills or mountains with large changes in elevation. The vegetation cover has been seriously damaged over the last century and, as a consequence, this area experiences geological hazards, such as landslides, debris and soil erosion during the rainy season (June–September). It is well recognised that rainfall redistribution from soil surface to deep soil layers influences the degree of soil erosion and probability of landslides. Forests and grasslands have more macropores compared with cropland, and also a higher resistance to surface runoff or soil erosion (Yang and Lu 2018). Previous dye-tracing experiments in this region showed that intensive human activities, such as deforestation and farming, can damage vegetation and reduce macroporosity and, therefore, preferential flow occurrences (Liu *et al.* 2018). However, a two-dimensional (2-D) or three-dimensional (3-D)

visualisation and quantitative analysis of macropore networks and their effects on preferential flow parameters is lacking in the Three Gorges mountain area.

Soil macropore structure is one of the most important factors controlling preferential flow. More precise parameterisation of macroporosity is needed to explain the relationship between soil macropores and preferential flow (Jarvis *et al.* 2016). Traditional methods for characterising soil structure consist of describing *in situ* morphology or thin sections of soil. However, these approaches cannot be used to obtain undisturbed and spatially continuous soil structure or macropore features at a high resolution with high efficiency. X-ray computed tomography techniques provide a viable method to reconstruct 3-D configurations of soil pedons without destroying their inner structures (Warner *et al.* 1989; Anderson *et al.* 1990; Gantzer and Anderson 2002; Cnudde and Boone 2013; Naveed *et al.* 2013). Better spatial resolution of pores (nm– μm) is usually associated with smaller sampling sizes (e.g. 1–10 mm in diameter) and, therefore, does not capture a pore scale suitable for quantitative studies of soil macropores. Medical computer tomography (CT) (μm –mm) is suitable and widely used in soil macropore structure studies, as it can analyse relatively large sample sizes (e.g. 50–200 mm in diameter) (Cnudde *et al.* 2006; Zhang *et al.* 2014). Parameters such as pore shape, diameter, direction and frequency distribution can be calculated through CT image analyses (Rab *et al.* 2014; Li *et al.* 2016). This method has been widely used to quantify 3-D soil macropore structures, in order to determine the relationship between soil pore characteristics and soil solute transport (Peyton *et al.* 1994; Heijs *et al.* 1995). Soil water infiltration and water redistribution in the soil profile are strongly affected by connectivity and tortuosity of soil macropores (Luo *et al.* 2008; Beven and Germann 2013). It is therefore valuable to detect preferential flow by characterising both soil macropore structure and corresponding water flow.

The direct methods for measuring preferential flow include dye tracing, combination of ponding infiltration and tension infiltration, and a breakthrough curve (BTC) method (Vervoort *et al.* 1999; Angulo-Jaramillo *et al.* 2000; Allaire *et al.* 2009). Dye trace experiments are destructive and, due to the influence of initial soil water content and underground water levels, field experiments with dye tracers and tension infiltrometers are variable with time and hard to control (Hardie *et al.* 2012; Zhang *et al.* 2014). The BTC method has been applied to detect preferential flow through intact soil cores in laboratory experiments with low cost equipment (Vervoort *et al.* 1999; Allaire *et al.* 2009). Compared with homogeneous soils, BTCs of soils with macropores have shorter solute penetration time and higher peak solute concentrations. The physical equilibrium model and physical non-equilibrium model combined with convection–dispersion equation (CDE) are often used to characterise homogenous and heterogeneous solute transport processes respectively. Combining the measured BTC data with these models allows determining the occurrence of preferential flow and its related solute flow parameters (Vervoort *et al.* 1999; Luo *et al.* 2010b). These soil hydraulic parameters are important for preferential water and solute flow modelling with hydrological models such as HYDRUS. Most previous

research on preferential flow focused on either the 3-D macropore properties (Hu *et al.* 2015; Zhang *et al.* 2017) or the characteristics of BTCs (Liu *et al.* 2017), with only a few studies focusing on both. Luo *et al.* (2010b) found that dispersivity was mainly affected by the macropore path number, hydraulic radius and mean angle, but this was in contrast to other research (Zhang *et al.* 2015). Also, the quantified relationships between other parameters (e.g. pore water velocity and solute dispersion coefficient) in the equilibrium or non-equilibrium CDE model and macropore properties are rarely reported.

The objectives of this study were to (1) characterise macropore structures under contrasting human disturbances (vegetation destruction, cultivation and trampling) using CT scanning and image analysis, (2) determine the corresponding preferential flow patterns by analysing the BTCs of nitrate in leaching experiments using the same undisturbed soil cores and (3) relate soil preferential flow to the soil macropore network and other soil properties.

Materials and methods

Study area

Field experiments were conducted in a mountain area (30°00'13"–31°28'30"N, 100°51'8"–111°39'30"E), which is located in the transitional zone of the upper- to middle-reaches of the Yangtze River. The local bedrocks consist of crystalline rocks from the middle-upper Proterozoic earthen base. According to meteorological records for 1994–2014 at a weather station in the study area (200 m elevation), the mean annual precipitation and evaporation were 1011 and 950 mm respectively. Rainfall intensity can be as high as 48 mm h⁻¹ in July–August, which contributes to serious soil erosion and flood hazard.

Experimental design and soil core sampling

The study area has experienced long-term destruction of native vegetation and intensive cultivation. Taking into consideration the intensity of human disturbance on vegetation and soil, three forest sites were selected along the hillslope with the elevation increasing from 600 to 1500 m (Fig. 1). Additionally, a deforested site in the middle-mountain area was included. The forest sites included Dalaoling (high forest, HF, 1450 m elevation), Baishuitou (middle forest, MF, 1187 m elevation) and Wangjialing (low forest, LF, 676 m elevation). The tea garden site (TG) is located at Dengcunping (1016 m elevation). The HF is located in the Dalaoling National Forest Park, and the vegetation and soil have experienced the lowest level of human disturbance. The dominant species (*Cryptomeria japonica*) were planted more than 40 years ago, and average stem diameter at breast height was 34 cm. The MF has experienced an intermediate level of human disturbance which includes cutting down big trees (stem diameter at breast height greater than 10 cm) five years before the start of the study, followed by natural reforestation dominated by secondary broadleaf forest and shrubs, such as *Quercus* and *Cunninghamia lanceolata*. The LF is characterised by intensive disturbance, since most fields were used as farmland or vegetable gardens until 10 years before the start of the

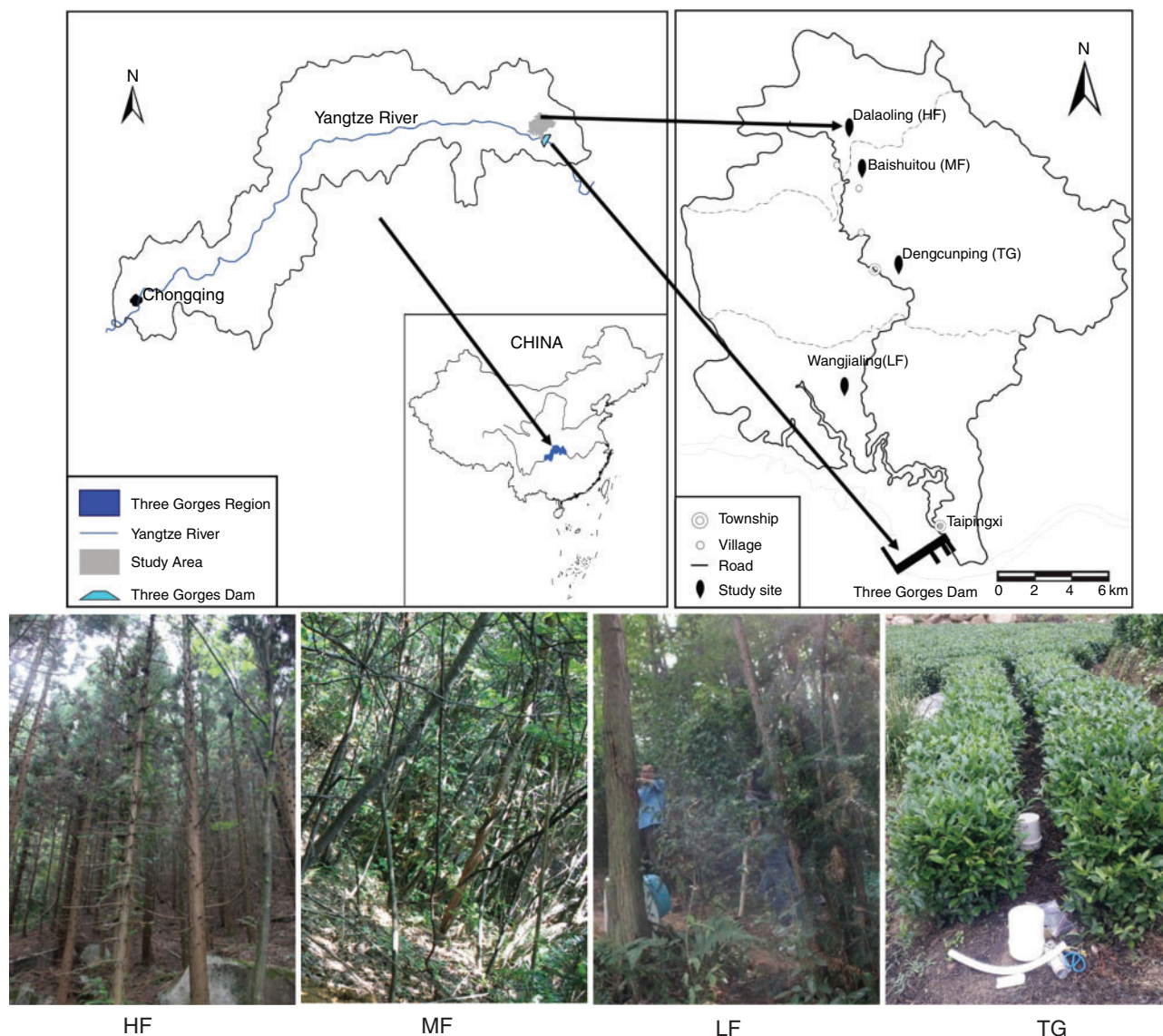


Fig. 1. Study area locations and representative vegetation types at the four study sites. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden.

experiment. Thereafter, coniferous trees were planted by the local government. At the time of the experiment, most of the original soil layers of the LF were lost due to soil erosion. Because tea is the main economic plant, the TGs typically cover much of the landscape, especially in the middle mountain areas. Before the tea bushes were planted, the original vegetation was cut down, their root systems were manually removed and the soil compacted by feet and sticks. The selected TG had been used as a tea garden for ~20 years before the start of the experiment, and is characterised by slight soil erosion due to terrace cultivation. Although more human disturbance has occurred in the TG than the LF in recent years, the intensity of human disturbance was relatively lower in the TG than the LF, as the soils were better protected (with thicker soil layers). In general, human disturbance level was ranked in a sequence of LF > TG > MF > HF.

After an initial investigation of the vegetation coverage at the four sites, eight locations at each site were selected to identify the typical soil distribution (0–40 cm) using soil samples collected with an auger and field observations (i.e. soil colour, soil texture and root distribution). Once the locations were confirmed, pits were excavated and three intact soil columns (representing three replicates) were collected with PVC tubes (10.4 cm in inner diameter and 40 cm in length) from each site. In total, 12 columns were prepared for both the CT scanning and the BTC experiments. Originally, we planned to obtain 40-cm-long intact soil columns from all sites, but this was difficult. The actual sampling lengths of soil columns for the HF, MF, TG, and LF were 30–35, 35–40, 30–40 and 22–25 cm respectively. Based on the soil profile observations, soil types of the HF, MF, TG and LF were identified (USDA classification) as Boralfs, Ustults, Ustults

and Udults respectively (Liu *et al.* 2018). Disturbed soil samples at each of the genetic soil horizons were collected for soil texture analysis.

Quantification of soil macropores using CT scanning

All undisturbed soil columns were scanned with a 64-row spiral medical CT scanner (AS+, Siemens, Germany) at Wuhan First Hospital. The peak voltage was adjusted to 130 V, current was 70 mA and resolution was 0.6 mm in both the vertical and horizontal directions. Cross-sections of soil columns were scanned starting from the soil surface and going downwards: 350–650 images were generated.

Following the definition of macropore given by Beven and Germann (1982), all pores identified by the CT scanner (≥ 0.6 mm) were considered as macropores. To reduce errors caused by possible disturbance, data from the top 1.0 cm of the soil columns (16 images) were excluded from the image analysis. For a balanced comparison of soil macropore parameters, soil CT images from the 1–20 cm depth were selected from each of the four study sites. To establish relationships between macropore features and the parameters of the BTCs, a circle with a diameter of 7.14 cm was selected in the centre of each image. After that, a threshold value of 40 was chosen by visually comparing the macropores in the binary images, using different threshold values and original images with the Fiji software (Schindelin *et al.* 2012), and this method was shown to be reliable (Li *et al.* 2016; Zhang *et al.* 2017).

Two-dimensional images were constructed with the 3D-Viewer plugin to produce 3-D macropore networks for each soil column (Schindelin *et al.* 2012). The 2-D macropore parameters of the macropore area and the macropore count along the soil depth were calculated with the continuous 2-D images, using the particle analysis tool in the Fiji software (Schindelin *et al.* 2012). All macropores were counted using this process. The macropore area and the macropore count of each 2-D image were divided by the sectional area (40 cm², equal to the area of resampled soil cores) to estimate macropore area density and macropore count density respectively. For the 3-D macropore parameter quantifications, the number, volume and surface area of all macropores ≥ 2 voxels (0.432 mm³) were measured with the 3-D objects counter tool. The 3-D skeleton of macropores was extracted and analysed with the Fiji boneJ plugin (Doube *et al.* 2010) to obtain the branch number, total branch length and branch Euclidean distance. Macroporosity, macropore surface area ratio, macropore number density, mean macropore size, length density, mean tortuosity, mean equivalent hydraulic radius and mean angle were calculated following Luo *et al.* (2010a). All 3-D soil macropore properties of the different soil layers (e.g. 1–7, 7–14 and 14–20 cm) were quantified with the same processes and were used to analyse their relationships with the BTC parameters.

Breakthrough curve

Previous BTC experiments with long soil columns (20 cm) demonstrated weak relationships between soil macroproperties and BTC parameters because of soil layering (Zhang *et al.* 2015). Therefore, the BTC experiments were carried out with

shorter soil cores (5 cm). The original longitudinal sections showed that soils adjacent to the inner edge of columns were disturbed. Hence, smaller undisturbed soil cores (7.14 cm in diameter) were resampled for BTC experiments. All soil columns were saturated by wetting them up from the bottom for 72 h, drained for 48 h to field capacity, before cutting them for BTC experiments. This helps to prevent soil disturbance during cutting. The 12 soil columns were cut vertically at 7 cm length intervals (for a total of 36 short soil columns). A cutting ring with a volume of 200 cm³ (5 cm high and 7.14 cm in inner diameter) was used to resample the soil core from the central part of each cut soil column, producing 36 resampled soil cores in total. The soil cores labelled as MF1–1, MF1–2 and MF1–3 represent soil cores from the MF1 soil column in depths of 1–7, 7–14 and 14–20 cm respectively. Soil cores from the other sites were labelled in the same way.

After re-saturation, saturated hydraulic conductivity of the 36 resampled soil cores was measured using the constant head method, and the BTCs of nitrate were obtained. Nitrate was selected as a tracer for the BTC experiments since it cannot be absorbed by the soil particles and is easily detected. Although it is reactive in natural soil profiles due to the action of soil microorganisms, it is stable in saturated conditions with purified water. First, the soil cores were rinsed with deionised water for 6–168 h until nitrate concentration in the outflow was <0.1 mg L⁻¹. A 20-mL nitrate solution with concentration of 10 mg L⁻¹ was applied to the top of the resampled soil cores in 3 s, and the nitrate application time was set as the initial time for the BTC parameters fitting. After the solute was completely infiltrated into the soil, deionised water was immediately added to the top of the resampled soil core and a constant water head of 1 cm was maintained. The outflow of the resampled soil core was collected at the bottom of the core with an interval of ~ 20 mL immediately after the nitrate was applied, and its weight was recorded. Afterwards, the nitrate concentrations of the outflow were determined with ultraviolet spectrophotometry (detection limit of 0.02 mg L⁻¹) until they were undetectable. The BTC measurement time had a range of 2–160 h because of differences in infiltration rates, and 40–50 outflow samples with the volume of ~ 20 mL were obtained for each BTC experiment.

After the experiment was completed, the nitrate-nitrogen content of the outflow solution was calculated. The BTC was found to be reliable and was used for further data analysis and model simulation, as long as the total amount of nitrate-nitrogen output was close to the amount of nitrate input (0.2 mg). The nitrate concentration versus flow rate was simulated using the CXTFIT software (Toride *et al.* 1995).

Under steady-state water flow conditions, one-dimensional transport of the solute in homogeneous soils can be described by the equilibrium CDE:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial z} \left(D \frac{\partial C}{\partial z} \right) - v \frac{\partial C}{\partial z} \quad (1)$$

where C is the solution concentration (mg L⁻³), t is time (h), z is the vertical distance (cm), D is the solute dispersion coefficient (cm² h⁻¹) and v is the mean pore water velocity (cm h⁻¹).

Assuming that molecular diffusion is ignored, the dispersivity, λ (cm), is given by Toride *et al.* (1995):

$$\lambda = \frac{D}{v} \quad (2)$$

As the equilibrium CDE equation is not appropriate to describe solute transport with preferential flows, the non-equilibrium model (e.g. mobile-immobile model or MIM) is usually used to describe solute transport in heterogeneous soils (Toride *et al.* 1995):

$$\theta_m \frac{\partial C_m}{\partial t} = \theta_m D_m \frac{\partial^2 C_m}{\partial z^2} - J_w \frac{\partial C_m}{\partial z} - \alpha (C_m - C_{im}) \quad (3)$$

where subscripts m and im denote the mobile and immobile domains respectively; θ is the soil water content; $J_w (= v_m \times \theta_m)$ is the water flux (cm h^{-1}), and α is the first-order mass transfer coefficient (h^{-1}). The α can be calculated using $\alpha = \omega \times J_w/L$, where ω is a dimensionless mass transfer coefficient and L is the length of the soil column (cm). The θ_m can be calculated as $\theta_m = \beta \times \theta$, where the dimensionless coefficient β represents the fraction of mobile water.

For the equilibrium CDE equation, v was fixed to be water flux divided by the saturated water content, while D was fitted. For the MIM model, parameters of D , β and ω were fitted with a known v . The D_m and v_m in the MIM model were then calculated using D , v and β (Zhang *et al.* 2015).

The use of the two-region model is not justified when $\beta = 1$ or $\omega = 100$. When $\beta = 1$, all soil pores are filled with mobile water, implying no differences in water flow. When $\omega = 100$, solute exchange between the mobile and immobile water is faster than the advection water flow, indicating that the solute transport is homogenous. In these two situations, the equilibrium CDE model rather than the MIM model is suitable to characterise BTCs. The effective dispersivity (λ_{eff}) – usually considered to be positively correlated with preferential density – was calculated as follows (Valocchi 1985):

$$\lambda_{\text{eff}} = \lambda_m \frac{\theta}{\theta_m} + q \frac{\left(1 - \frac{\theta_m}{\theta}\right)^2}{\alpha} \quad (4)$$

where q is water flux (cm h^{-1}).

Determination of other soil properties

To obtain more reliable relationships between preferential flow characteristics and soil physical properties, including soil macropore structures, saturated water content (θ_s), saturated hydraulic conductivity (K_{sat}), soil organic matter (SOM) content, bulk density (BD) and root length density (RLD) were measured from the same resampled 5-cm-long soil cores.

After the BTC experiment was completed, θ_s was measured immediately and soil BD determined with the drying method (105°C). The dried soil samples were crushed to separate the plant roots and gravel from the soil. The cleaned roots were scanned with a scanner (Huipu 7500, USA) to determine the RLD using the Fiji software. The soil samples without roots or gravel were passed through a mesh size of 100 (0.149 mm) to measure the SOM content using the potassium dichromate oxidation method (Walkley and Black 1934). The relevant soil properties are shown in Table 1.

Statistical analysis

A one-way analysis of variance combined with least significant difference ($P < 0.05$) test was used to test differences in the means of the soil macropore structure and the BTC parameters. Pearson correlation analysis was conducted among the macropore parameters (i.e. macroporosity, macropore number per volume, mean macropore size, mean angle, length density, mean tortuosity, mean equivalent hydraulic radius and node density), BTC parameters (v , D , v_m , D_m , λ , λ_m and λ_{eff}), soil properties (SOM content, BD, θ_s and K_{sat}) and RLD. Some macropore properties (i.e. surface area density, mean macropore size, length density and node density) and BTC parameters (v , D , v_m and λ) were log-transformed because they were not normally distributed. Statistical software SPSS 22.0 was used for all related analyses.

Results

Visualisation of macropores

The original longitudinal sections and produced 3-D macropore maps of the undisturbed soil columns are shown in Fig. 2 and Fig. 3 respectively. In the original scanned images, the black, grey and white patches always represent macropores or cracks,

Table 1. Soil physical properties at the four study sites

A, humus horizon; Ap, the mineral horizon formed below O horizon in cultivated land; E, eluvial horizon; B, illuviation horizon; C, parent material horizon; SOM, soil organic matter; K_{sat} , saturated hydraulic conductivity; θ_s , saturated water content; RLD, root length density; HF, high elevation forest; MF, mid-elevation forest; LF, low elevation forest; TG, tea garden

Sites	Soil depth (cm)	Bulk density (g cm^{-3})	SOM content (g kg^{-1})	Sand	Soil texture Silt (%)	Clay	K_{sat} (cm h^{-1})	θ_s ($\text{cm}^3 \text{cm}^{-3}$)	RLD (cm cm^{-3})
HF	0–15 (A)	0.87	51.51	10.82	57.31	31.86	69.86	0.54	0.94
	15–30 (E)	1.00	37.42	14.29	52.64	33.07	36.35	0.52	0.96
MF	0–10 (A)	0.81	54.24	22.14	48.16	29.69	88.08	0.54	0.92
	10–20 (E)	0.88	36.63	19.42	52.48	28.10	48.36	0.53	0.75
TG	20–40 (B)	0.94	22.75	19.51	50.62	29.87	15.84	0.53	0.55
	0–12 (Ap)	1.08	33.51	19.58	66.83	13.59	24.67	0.51	0.44
LF	12–40 (E)	1.27	19.08	26.27	49.90	23.83	4.20	0.49	0.13
	0–12 (A)	1.18	27.99	69.91	15.52	15.57	12.44	0.49	0.30
	12–30 (C)	1.43	8.79	84.80	8.82	6.38	2.09	0.43	0.04

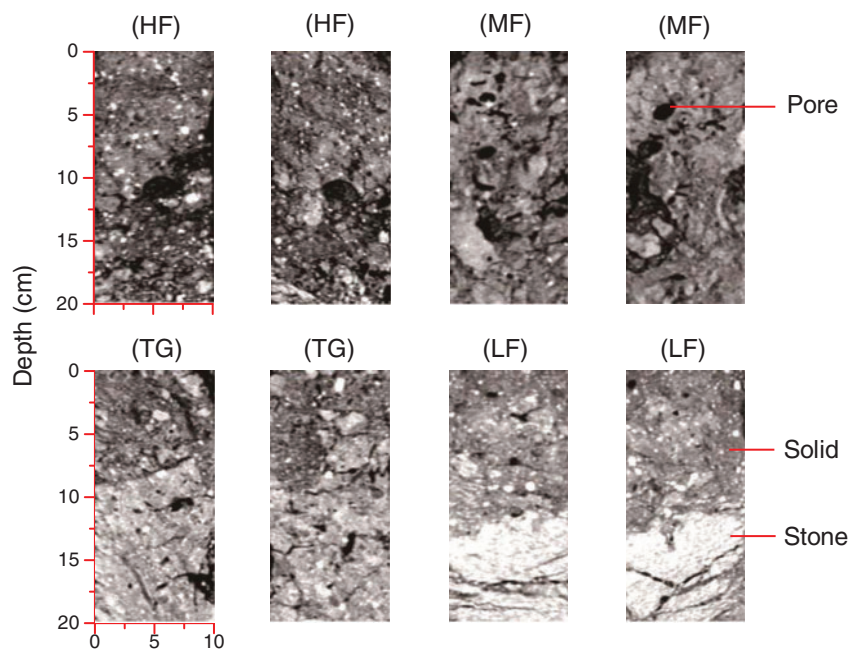


Fig. 2. Longitudinal cross-sections of the same soil column in two orthogonal directions at different sites (scale: cm). Macropores are shown in black, solid particles in grey and stones in white. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden.

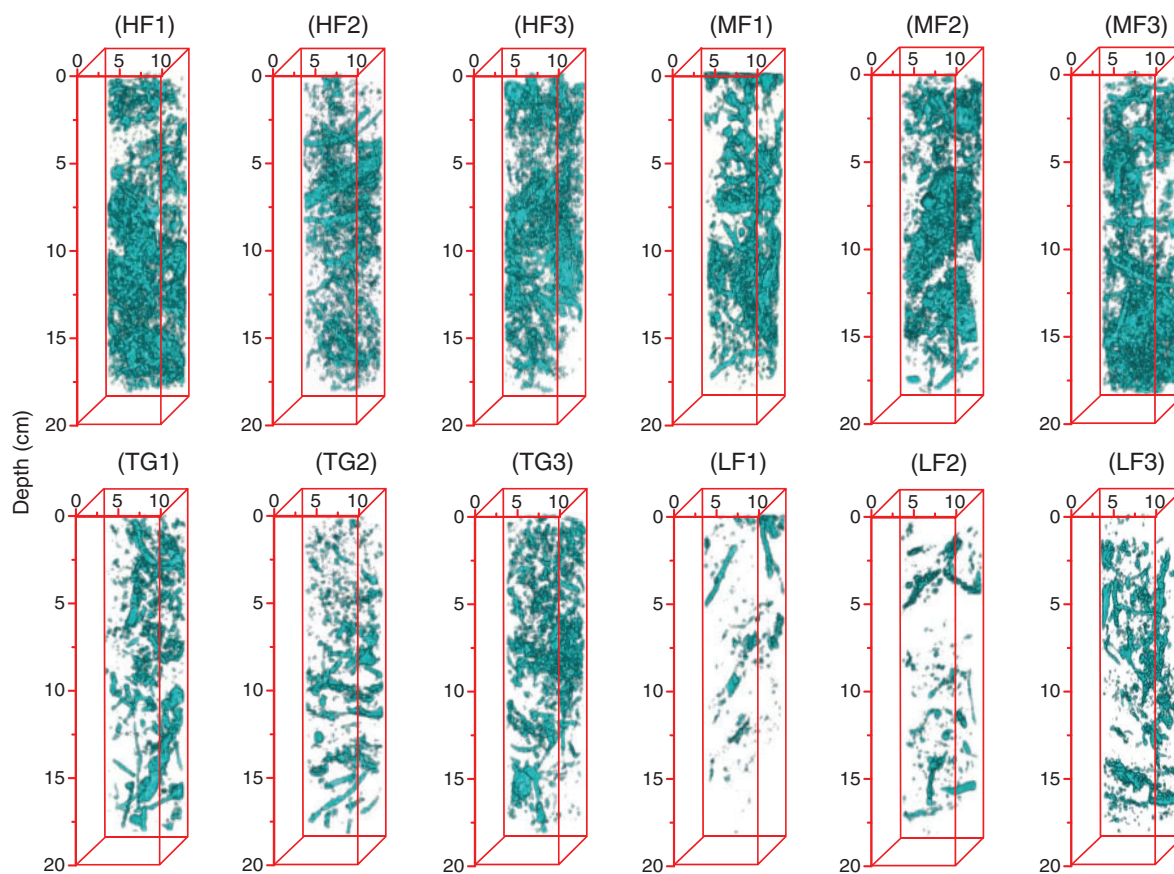


Fig. 3. Three-dimensional images of the soil columns at different sites (scale: cm). Macropores are shown in blue and solid particles in white in the three-dimensional images. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden. Numbers 1–3 represent the replicates.

soil matrix and gravel or rock respectively (Li *et al.* 2016). The 2-D images (Fig. 2) showed considerable differences in the size, shape and position of macropores between the four sites or among different soil depths for the same site. Generally, HF and MF had more macropores than TG, followed by LF, but the gravel content of TG and LF was higher than that of HF and MF. Two obviously different soil horizons were easily identified in the TG and LF sites, whereas soils from the HF and MF sites were more homogenous. The 3-D reconstructed macropore structures (Fig. 3) were consistent with the 2-D observations. The macropores in HF and MF were more abundant compared with TG and LF, and their morphology was more complex. Except for one soil column (LF3) from the LF site (Fig. 3), fewer macropores were identified in LF than in TG.

Macroporosity, surface area density and macropore area density

The mean macroporosities from the 0–20 cm soil depth of HF, MF, TG and LF were 0.115, 0.111, 0.037 and 0.015 m³ m⁻³ respectively (Table 2). Statistical tests showed that the macroporosities of HF and MF were significantly higher than those of TG and LF, while no significant differences were observed between HF and MF, or between TG and LF. In addition, higher standard deviation of macroporosity was observed in HF than the other three sites. The macropore surface area densities varied between the four study sites in a way similar to that of macroporosity (Table 2) – with higher values in HF and MF than in TG and LF.

There were slight differences in the macropore area densities between the three soil columns extracted from the same sites (Fig. 4). This indicated a high degree of heterogeneity and randomness of macropores for the forest soils. The range of macropore area densities throughout the depth was largest at the HF site, followed by the MF, TG and LF, in the same

descending order as the macroporosity and macropore surface area.

Macropore number density, length density and count density

The macropore number densities of HF, MF, TG and LF were 20.74×10^5 , 14.79×10^5 , 11.28×10^5 and 7.40×10^5 N m⁻³ respectively (Table 2). A higher macropore number density was observed in HF than in the TG and LF, with no significant differences between HF and MF, or between MF and TG or LF. Similarly, significantly higher macropore length densities were observed in HF and MF than in TG and LF (Table 2). No significant differences in the macropore length densities were identified between HF and MF, or between TG and LF. Differences in macropore count density were also observed with soil depth in the four sites (Fig. 5). At the HF site, macropore count density increased gradually from soil surface to 5 cm, and fluctuated around 1.10 cm⁻² for 5–20 cm depth. The macropore count densities at the MF and LF were comparable and fluctuated around 0.75 and 0.25 cm⁻² respectively. In contrast, the macropore count densities at TG decreased continuously with soil depth (Fig. 5).

Macropore size distribution, mean macropore size and hydraulic radius

The macropore size distribution in different soil columns is shown in Fig. 6. For HF and MF, the macropore size distributions were similar among replicates for volumes <10 000 mm³, but the TG soil only showed similar size distribution for volumes <1000 mm³. In contrast, no significant differences in macropore size distribution were observed between the three replicate soil columns in LF. Most of the macropores were <10 000 mm³, but the larger macropores in HF, MF and LF occupied a vast majority of the total macropore volume. The mean macropore sizes at the HF, MF, TG and LF sites were 62.24,

Table 2. Statistical summary of macropore properties for the whole soil column from the four study sites

HF, high elevation forest; MF, mid-elevation forest; LF, low elevation forest; TG, tea garden. Means within the same column that do not share a letter statistically differ at $P < 0.05$

Soil column	Macroporosity (m ³ m ⁻³)	Surface area density (m ² m ⁻³)	Mean macropore size (mm ³)	Macropore number density ($\times 10^5$ N m ⁻³)	Mean angle (°)	Length density (km m ⁻³)	Mean tortuosity	Hydraulic radius (mm)	Node density ($\times 10^6$ N m ⁻³)
HF1	0.143	313.66	81.71	17.51	27.07	43.69	1.42	1.02	45.65
HF2	0.085	255.25	29.01	29.33	27.39	27.03	1.53	1.00	26.14
HF3	0.117	269.38	76.01	15.38	28.96	32.69	1.58	1.07	29.39
Mean	0.115 (0.024) a	279.43 (24.88) a	62.24 (23.61) a	20.74 (6.14) a	27.81 (0.82) a	34.47 (6.92) a	1.51 (0.07) a	1.03 (0.03) b	33.73 (8.54) a
MF1	0.100	99.68	79.23	12.58	26.38	18.36	1.44	1.32	18.78
MF2	0.116	251.86	75.61	15.36	27.71	33.39	1.50	1.05	27.79
MF3	0.118	265.53	72.00	16.44	28.74	37.03	1.51	1.01	33.39
Mean	0.111 (0.008) a	205.69 (75.17) a	75.61 (2.95) a	14.79 (1.62) ab	27.61 (0.96) a	29.59 (8.08) a	1.48 (0.03) a	1.13 (0.14) b	26.65 (6.02) a
TG1	0.042	100.01	38.12	11.01	26.20	5.59	1.47	1.55	6.59
TG2	0.019	52.36	21.56	8.71	17.64	1.91	1.43	1.77	2.97
TG3	0.049	129.57	35.00	14.12	25.78	8.17	1.52	1.39	8.91
Mean	0.037 (0.013) b	93.98 (31.81) b	31.56 (7.19) b	11.28 (2.22) b	23.21 (3.94) a	5.23 (2.57) b	1.47 (0.04) a	1.57 (0.16) a	6.16 (2.44) b
LF1	0.007	21.88	15.65	4.53	24.61	0.61	1.73	1.93	1.24
LF2	0.012	35.56	19.24	6.19	19.26	1.52	1.49	1.58	2.49
LF3	0.026	77.01	22.57	11.47	26.75	4.45	1.41	1.36	6.36
Mean	0.015 (0.008) b	44.82 (23.44) b	19.15 (2.83) b	7.40 (2.96) b	23.54 (3.15) a	2.19 (1.64) b	1.54 (0.14) a	1.62 (0.23) a	3.36 (2.18) b

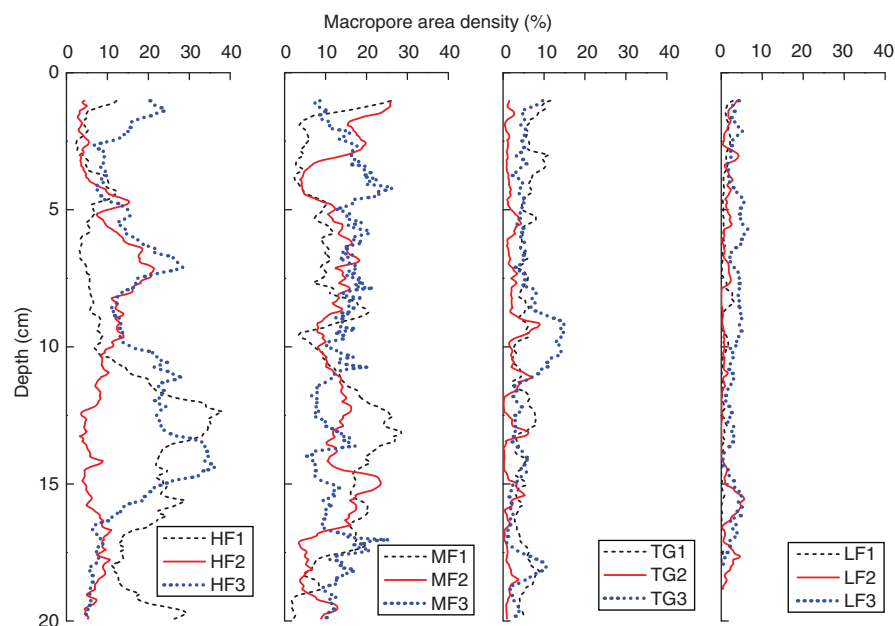


Fig. 4. Macropore area density distribution with soil depth at different sites. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden. Numbers 1–3 represent the replicates.

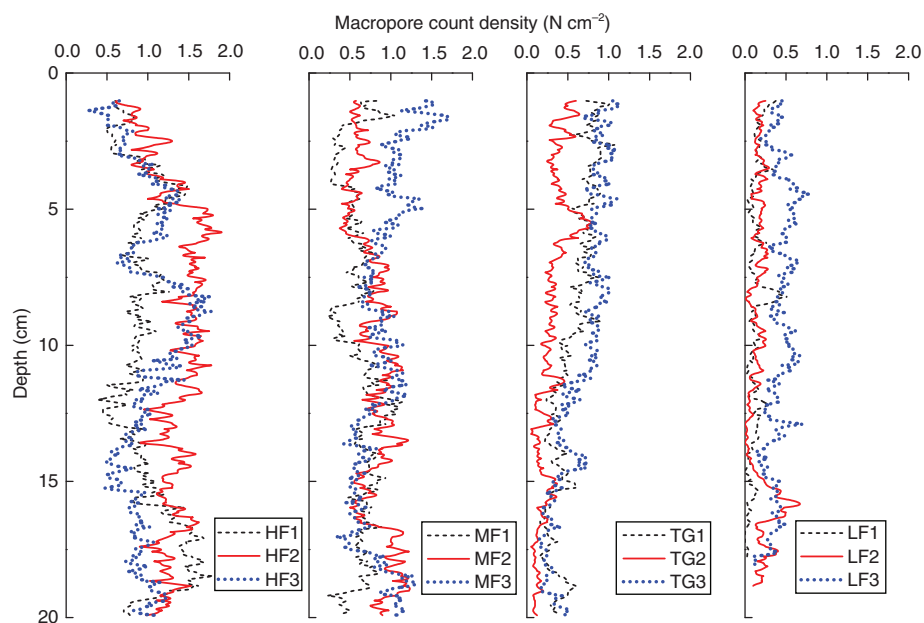


Fig. 5. Macropore count density distribution along soil depth at different sample sites. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden. Numbers 1–3 represent the replicates.

75.61, 31.56 and 19.15 mm³ respectively (Table 2). No significant differences in macropore sizes were observed between HF and MF, or between TG and LF. However, TG and LF sites had significantly smaller macropore sizes compared with HF and MF. In contrast to the rank of mean macropore sizes, larger hydraulic radii were identified in TG (1.57 mm) and LF (1.62 mm) sites than in HF (1.03 mm) and MF (1.13 mm) sites. No significant

differences in hydraulic radii were observed between HF and MF, or between TG and LF.

Macropore tortuosity, inclination and connectivity

Higher tortuosity means that more time is needed for water and solute to pass through soil macropores. No significant

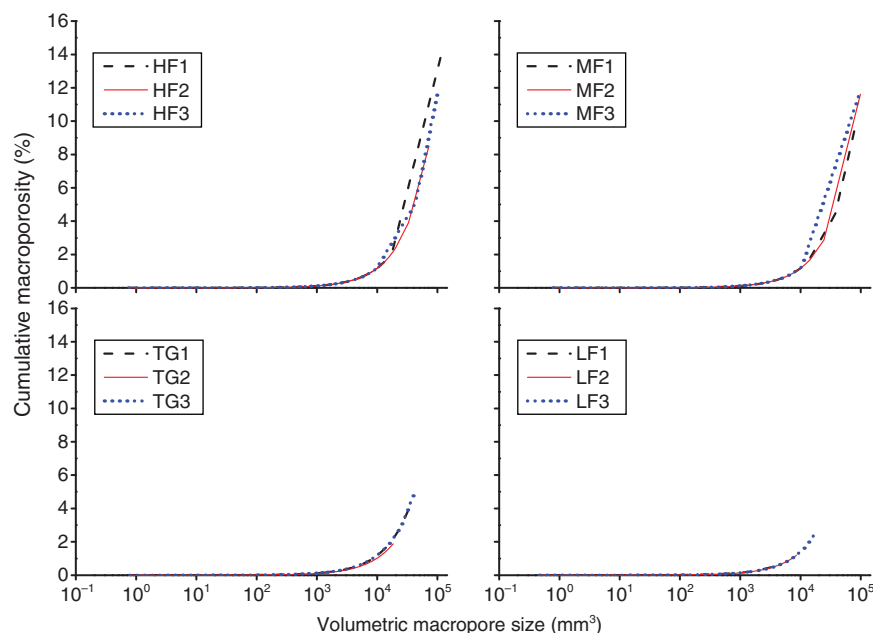


Fig. 6. Macropore size distribution (by volume) in each of the soil columns investigated at the four study sites. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden.

differences in mean tortuosity and mean angle were observed among the four study sites. The mean tortuosities for HF, MF, TG and LF sites were 1.51, 1.48, 1.47 and 1.54 respectively. A large mean angle indicates that the soil macropores are more vertically oriented. Mean angles for HF, MF, TG and LF sites were 27.81° , 27.61° , 23.21° and 23.54° respectively. Soil macropore connectivity can be presented by the node density for the soil column and larger scales (cm or larger) (Luo *et al.* 2010a). A larger node density indicates a higher connection within the macropore network. Although no significant differences in macropore tortuosity and inclination were observed between the four sites, higher node densities were identified in HF and MF than in TG and LF.

Breakthrough curve parameters

A selected set of experimental observations from a BTC and the associated fitting curves using CDE or MIM models at the four sites is shown in Fig. 7. Nitrate penetrated quickly in most soil cores, as nitrate in the outflow was detected at 0.1 pore volume for most soil cores. There were many BTCs with asymmetry and pronounced tailing, indicating the existence of preferential flow. The peak values of the relative nitrate concentration and their corresponding outflow times for the BTCs ranged within 0.090–0.184 and from 0.154–1.109 pore volumes respectively. Among all samples, the maximum peak of relative concentration (0.184) was observed in the 1–7 cm layer in the HF3 soil column (HF3–1), and the minimum peak (0.090) was observed at 14–20 cm layer in the MF2 column (MF2–3, data not shown).

Fitting parameters derived from the equilibrium CDE and MIM models are shown in Table 3. Although the CDE can be used to describe the BTCs for all soil cores, some BTCs better fitted the MIM model than the equilibrium CDE model

according to the higher coefficients of determination for the MIM model (adjusted $R^2 = 0.82$ – 1.00) than those for the equilibrium CDE model (adjusted $R^2 = 0.73$ – 0.99). The BTCs of most soil columns in HF and MF could be simulated well with both CDE and MIM models, except for HF1–2. In contrast, about half of the BTCs in the TG (TG2–1, TG2–2, TG3–2 and TG3–3) and LF sites (LF1–2, LF2–1, LF2–3 and LF3–2) did not fit well for the MIM model, indicating that preferential flow was weaker in TG and LF compared with HF and MF.

No significant differences in terms of goodness-of-fit with the equilibrium CDE equation were observed between the four study sites (Table 3). Higher ν and D values (mean values of whole soil columns were compared for all BTC parameters) were observed at the MF than at the TG and LF sites, whereas no significant differences in ν and D values were observed between HF, TG and LF sites. In addition, no significant differences in λ were observed among the four sites. No significant differences in terms of goodness-of-fit with the MIM and MIM parameters β , ω , α , λ_m and λ_{eff} were identified among the four sites (Table 3), although these parameters were slightly higher in the HF, MF and TG sites than LF. Higher ν_m values were identified in MF than LF, while no significant differences were observed between HF and MF, and between TG and LF. Lower D_m values were observed at the LF than at the other three sites, but there were no significant differences in D_m values among HF, MF and TG sites.

Discussion

Effects of root distribution and soil physical properties on macropore parameters

Soil physical properties (e.g. SOM, soil texture and BD) change slowly under the influence of precipitation and temperature under natural conditions. However, human disturbances

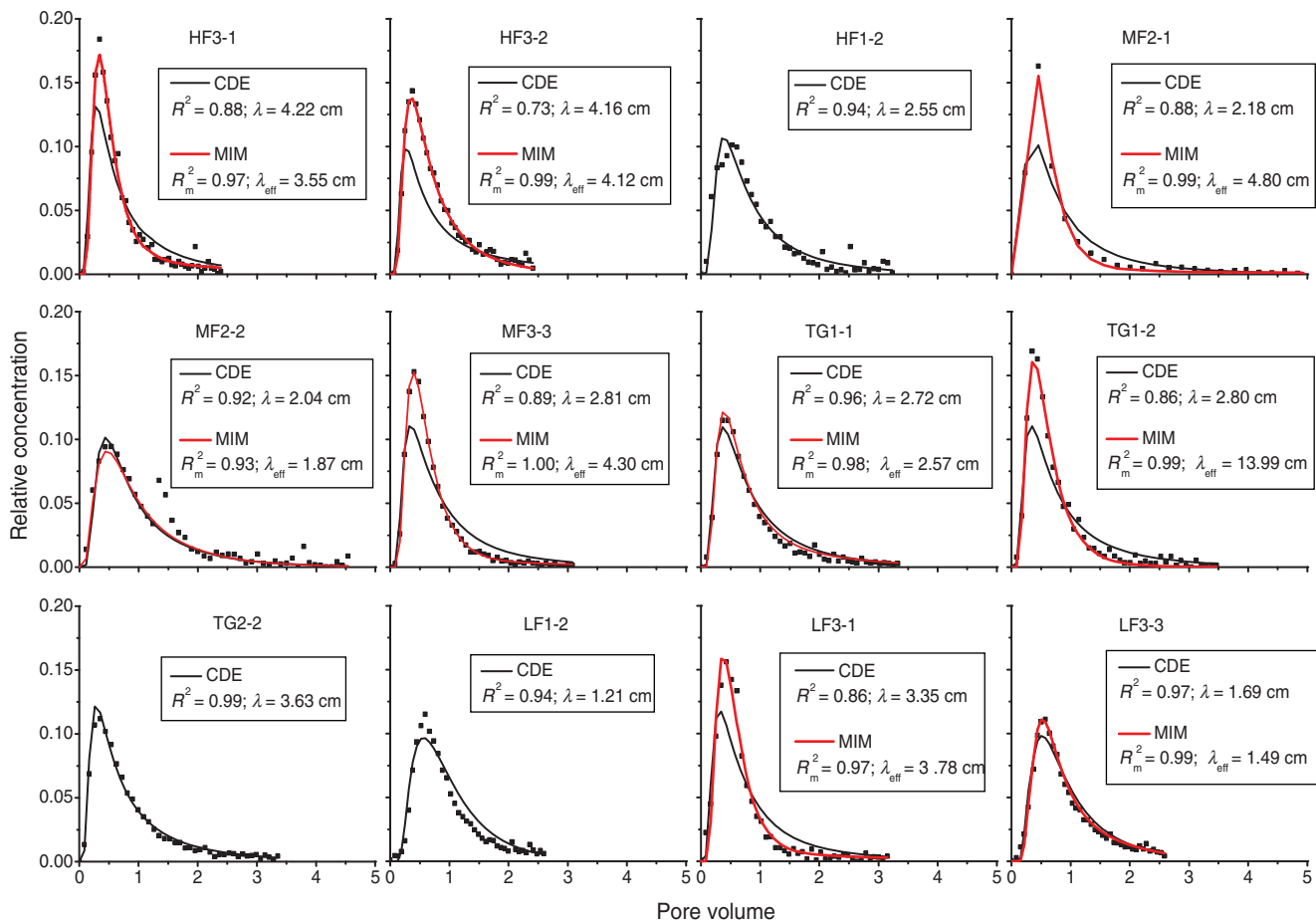


Fig. 7. Representative graphs of observed and fitted breakthrough curves by the equilibrium CDE and MIM models for the soil columns from the four sites. HF is high elevation forest, MF is mid-elevation forest, LF is low elevation forest and TG is tea garden. The first and second numbers (separated by a hyphen) represent replicates (1–3) and depth (1 represents 1–7 cm, 2 represents 7–14 cm and 3 represents 14–20 cm) respectively.

(i.e. vegetation destruction, cultivation and trampling) caused rapid changes in these soil properties and affect soil macropores. Although the variability of some soil properties was ascribed to natural spatial variations, considerable differences were apparent. In the four study sites, less human disturbance corresponded to lower soil BD, higher SOM content, higher clay content and higher RLD (Table 1), which resulted in more soil macropores.

Soil macropores are greatly affected by root systems, which are closely related to the degree of vegetation destruction, vegetation type and growth year. Root expansion and penetration can affect soil BD and porosity adjacent to the root. Vegetation growth can also increase soil organic carbon stocks, which improves soil structure and macropore development (Beven and Germann 1982). According to Laine-Kaulio *et al.* (2015), root channels can serve as stable macropores long after root decay. Significant positive relationships were observed between RLD and macroporosity, surface area density, mean macropore size, length density and node density, but negative relationships were found between RLD and mean equivalent hydraulic radius (Table 4). This shows that plant root systems improve soil structure, especially macropore properties.

This is in agreement with Seobi *et al.* (2005), who attributed greater porosity at 0–30 cm soil depth in an agroforestry buffer zone compared with the crop row to the greater root density abundance. Similarly, Hu *et al.* (2015) observed more soil macropores immediately under the shrub canopy than the surrounding areas. More macropores are always observed in soils with greater root densities (Udawatta *et al.* 2006). However, smaller macropore hydraulic radii in HF and MF were attributed to an abundance of fine root systems, which corresponded with higher RLD in these sites.

Other than root distribution, soil texture also affects the occurrence of macropores, because sandy soils usually have larger soil pores than loamy and clay soils. The highest sand content was measured in the LF, and resulted from serious deforestation and soil erosion over the 10 years before the start of this experiment. This might have restricted the macropore development at this site. Normally, a positive relationship is found between sand content and K_{sat} (Qiao *et al.* 2018); however, the lowest K_{sat} was observed at the LF site, which had the highest sand content of the four sites (Table 1). This was due to fewer macropores at the LF site compared with the other sites. According to Kohne *et al.* (2009), relatively high clay

Table 3. Fitting parameters derived by the equilibrium convection–dispersion equation (CDE) and the two-region mobile–immobile model (MIM)

The fitted parameters were invalid for $\beta = 1$ or $\omega = 100$ and excluded for calculating mean values and comparing the difference between treatments. HF, high elevation forest; MF, mid-elevation forest; LF, low elevation forest; TG, tea garden; v , average pore water velocity; D , solute dispersion coefficient; λ , dispersivity; v_m , average pore water velocity of the mobile region; D_m , solute dispersion coefficient of the mobile region; β , fraction of mobile water; ω , dimensionless mass transfer coefficient; λ_m , dispersivity of the mobile region; λ_{eff} , effective dispersivity; α , first-order mass transfer coefficient; s.d., standard deviation. Means within the same column that do not share a letter statistically differ at $P < 0.05$

Site	Value	Equilibrium CDE						MIM					
		v (cm min ⁻¹)	D (cm ² min ⁻¹)	λ (cm)	R^2	v_m (cm min ⁻¹)	D_m (cm ² min ⁻¹)	β	ω	λ_m (cm)	λ_{eff} (cm)	α	R_m^2
HF	Max	1.24	3.13	4.22	0.96	2.36	1.18	0.76	24.30	1.49	4.12	8.54	1.00
	Median	0.34	0.58	2.35	0.90	0.76	0.36	0.61	0.35	0.52	2.48	0.05	0.98
	Min	0.01	0.02	0.56	0.73	0.29	0.11	0.17	0.10	0.11	0.35	0.02	0.82
	Mean	0.42 ^{ab}	0.97 ^{ab}	2.40 ^a	0.89 ^a	0.98 ^{ab}	0.46 ^a	0.57 ^a	3.67 ^a	0.63 ^a	2.36 ^a	1.18 ^a	0.96 ^a
	s.d.	0.33	0.96	1.07	0.07	0.73	0.36	0.19	8.39	0.49	1.26	2.98	0.05
MF	Max	2.64	6.16	3.75	0.98	3.98	3.17	0.87	61.88	1.08	5.22	18.77	1.00
	Median	0.79	2.04	2.28	0.93	1.52	0.69	0.58	0.16	0.80	2.76	0.11	0.99
	Min	0.26	0.40	1.39	0.88	0.33	0.26	0.20	0.04	0.35	1.79	0.00	0.93
	Mean	0.99 ^a	2.52 ^a	2.41 ^a	0.93 ^a	1.73 ^a	1.19 ^a	0.60 ^a	7.96 ^a	0.73 ^a	3.19 ^a	2.48 ^a	0.98 ^a
	s.d.	0.87	2.29	0.68	0.04	1.28	0.93	0.17	19.23	0.23	1.31	5.82	0.02
TG	Max	0.59	1.13	3.63	0.99	0.70	0.83	0.84	14.95	1.18	13.99	0.76	0.99
	Median	0.08	0.28	2.72	0.95	0.37	0.23	0.71	0.12	0.66	2.57	0.01	0.98
	Min	0.02	0.05	1.44	0.83	0.07	0.08	0.30	0.04	0.31	0.16	0.00	0.96
	Mean	0.17 ^b	0.38 ^b	2.54 ^a	0.92 ^a	0.37 ^{ab}	0.30 ^a	0.63 ^a	3.10 ^a	0.77 ^a	4.71 ^a	0.16 ^a	0.97 ^a
	s.d.	0.18	0.35	0.69	0.05	0.26	0.30	0.32	6.06	0.45	5.08	0.31	0.40
LF	Max	0.20	0.38	3.35	0.98	0.38	0.09	0.71	0.43	0.58	3.78	0.03	1.00
	Median	0.08	0.20	1.77	0.94	0.15	0.08	0.53	0.34	0.20	1.49	0.01	0.99
	Min	0.02	0.02	1.10	0.79	0.12	0.07	0.53	0.21	0.01	0.19	0.01	0.97
	Mean	0.08 ^b	0.19 ^b	2.09 ^a	0.92 ^a	0.16 ^b	0.06 ^b	0.44 ^a	0.24 ^a	0.33 ^a	1.35 ^a	0.01 ^a	0.99 ^a
	s.d.	0.07	0.15	1.01	0.06	0.14	0.01	0.10	0.11	0.21	1.18	0.01	0.01

Table 4. Pearson correlation coefficient (R) between different soil physical properties, macropore properties as determined from CT images and solute transport parameters fitted by the BTCs

BD, bulk density; SOM, soil organic matter content; RLD, root length density; Ma, macroporosity; SAD, surface area density; MMS, mean macropore size; MND, macropore number density; $\bar{\theta}$, mean angle; LD, length density; $\bar{\tau}$, mean tortuosity; $\bar{r}$, mean equivalent hydraulic radius; ND, node density; v , average pore water velocity; D , solute dispersion coefficient; v_m , average pore water velocity of the mobile region; D_m , solute dispersion coefficient of the mobile region; λ , dispersivity; λ_m , dispersivity of the mobile region. *, significant at $P < 0.05$; **, significant at $P < 0.01$.

	BD	SOM	RLD	Ma	SAD	MMS	$\bar{\theta}$	LD	$\bar{\tau}$	$\bar{r}$	MND	v	D	v_m	D_m	λ	λ_m
SOM	-0.85**																
RLD	-0.72**	0.74**															
Ma	-0.72**	0.53**	0.73**														
SAD	-0.80**	0.59**	0.75**	0.87**													
MMS	-0.49**	0.25	0.50*	0.76**	0.65**												
$\bar{\theta}$	-0.42**	0.38*	0.27	0.26	0.27	0.22											
LD	-0.73**	0.56*	0.73**	0.85**	0.95*	0.69	0.26										
$\bar{\tau}$	0.27	-0.08	-0.35	-0.34*	-0.52**	-0.58**	-0.05	-0.57**									
$\bar{r}$	0.53**	-0.36*	-0.53**	-0.62**	-0.80**	-0.48**	-0.21	-0.84**	0.74**								
MND	-0.74**	0.61*	0.78**	0.88*	0.93**	0.63**	0.25	0.98**	-0.45*	-0.76**							
v	-0.83**	0.68**	0.71*	0.70**	0.71**	0.49**	0.33	0.62**	-0.15	-0.33	0.63**						
D	-0.80**	0.69**	0.74*	0.71**	0.69**	0.56**	0.33	0.62**	-0.17	-0.32	0.63**	0.96**					
v_m	-0.86**	0.74**	0.74**	0.72**	0.71**	0.50**	0.29	0.62**	-0.20	-0.31	0.64**	0.92**	0.86**				
D_m	-0.57**	0.52**	0.49*	0.70**	0.55**	0.58*	0.16	0.52*	-0.17	-0.31	0.53**	0.74**	0.75**	0.68**			
λ	-0.17	0.17	0.05	-0.05	-0.12	0.03	0.26	-0.29	0.34	0.42**	-0.30	0.53**	0.55**	0.38*	0.36*		
λ_m	0.02	0.04	0.16	0.14	0.14	0.27	0.04	0.16	0.03	0.07	0.14	0.01	0.14	-0.24	0.16	0.19	
λ_{eff}	-0.06	0.06	0.28	-0.01	0.09	0.17	0.11	0.08	-0.09	0.05	0.04	0.25	0.36*	0.10	0.11	0.40*	0.25

content is essential for the aggregate formation and the conservation of stable macropore structures. The high sand content at the LF site does not promote stable macropore formation (Zhang *et al.* 2017). Sandy soils tend to be less

fertile than loamy soils, which would restrict fauna activities and biopore formation (Luo *et al.* 2010b).

Macropore formation is also affected by farming practices. Luo *et al.* (2010a) identified differences in macropores

between pasture and farmland. They attributed the reduced macroporosity in farmland to farming practices which caused compaction and decreased root growth. In our case, the TG sites showed a higher BD than the forest sites (HF and MF). The lower macroporosity at the TG likely resulted from 20 years of tea cultivation. Before the tea bush planting, the original vegetation was cut down and root systems removed, and this would have compacted the soil. Although the macroporosity can be restored by root penetration of the tea bush into soil, continuous human trampling during cultivation and tea leaf-picking can result in high BD and reduced macroporosity, which counteracts the plant effects. This is supported by the significant negative relationship between soil BD and many soil macropore parameters (i.e. macroporosity, surface area density, mean macropore size, angle, length density and macropore number density).

Macropore distribution can be effected by SOM content, which is essential for stable aggregate formation (Zhang *et al.* 2017). Significant positive relationships were identified between SOM content and macroporosity, macropore surface area density, mean angle, length density and node density, but there was a negative relationship between SOM content and macropore hydraulic radius (Table 4). Thus, SOM enhanced both the formation of smaller macropores and also the stabilising of existing bigger macropores, and increased macropore length. Finally, the lower SOM content of the TG and LF sites, as a result of intensive human disturbances, may also have reduced the number of macropores.

Effects of macropores on BTCs and preferential flow

Preferential flow was widely observed at the four sites, as indicated by the fast breakthrough, long tails and asymmetry of the BTCs. The equilibrium CDE model is also suitable for non-equilibrium solute transport modelling if the preferential flow is not particularly strong (Luo *et al.* 2010a). In our study, the MIM worked better than the equilibrium CDE equation (Table 2). This was probably because the nitrate transport at the HF and MF sites followed a preferential flow, as a result of greater macroporosity, macropore surface area and mean macropore size. In contrast, the MIM model was not suitable for fitting every BTC, and was especially poor at the TG and LF sites, indicating that homogenous water flow was more likely at these sites. Neither the MIM nor the equilibrium CDE models fitted the BTC characterised by a double solute concentration peak (i.e. MF2–2), indicating that a dual permeability model should be considered in the future (Liu *et al.* 2017).

The positive relationships of v (v_m) or D (D_m) with macroporosity, macropore surface density, mean macropore size, length density and macropore number density (Table 3) indicated that the BTCs were affected by macropores, in agreement with other results (Zhang *et al.* 2015). In contrast, no significant relationships were observed between v (v_m) or D (D_m) and macropore mean angle, mean tortuosity and mean hydraulic radius. This can be attributed to many unconnected small macropores in soil cores, which mainly exist between soil aggregates.

Peak concentrations of outflow (C_{peak}) were affected by macropore characterisation. Although no significant

relationships were observed between C_{peak} and macropore parameters (data not shown), higher C_{peak} values were usually observed in the HF and MF sites, where more macropores were observed compared with the other two sites (Fig. 7). This could be explained by faster water and solute transport through the macropores than through the soil matrix, which would reduce the breakthrough time and enhance C_{peak} (Luo *et al.* 2010a). The λ (λ_{eff}) was used to describe the intensity of preferential flow, and stronger preferential flow always corresponded to higher λ (λ_{eff}) values (Vervoort *et al.* 1999). For example, Luo *et al.* (2010b) observed a higher λ (λ_{eff}) in a soil column with a long earthworm hole. Although no significant differences in λ (λ_{eff}) were observed between the four study sites, slightly higher λ (λ_{eff}) were observed in HF, MF and TG than in LF (Table 3). Luo *et al.* (2010b) suggested that a sample diameter of 10 cm was too small to obtain reliable λ measurements for soils with abundant macropores. Thus, the small soil cores (7 cm in diameter) in this study could be responsible for the non-significant differences in λ values among sites. Also, poor relationships were observed between λ (λ_{eff}) and all macropore parameters in our study, except for a significant positive relationship between equivalent hydraulic radius and λ . This implies that the radius was more important for flow processes than other macropore parameters used in this study. The possible reason for these poor relationships is that some soil macropores in our study were unconnected and made minimal contributions to preferential water flow. This poor relationship between macropore porosities and λ (λ_{eff}) was also reported by Zhang *et al.* (2015), who attributed it to large variations in soil macropore properties throughout the soil column. Additionally, the BTCs and their relevant parameters will also be affected by the macropore type (e.g. biopores or non-biopores, percolating pores and inter-aggregates pores), which will affect hydraulic parameters differently (Zhou *et al.* 2017). Differences in macropore types were not considered in this study, but were likely one reason for poor relationships between macropore porosities and BTC parameters. In addition, undetected macropores (<0.6 mm) were also important for water and solute transport, and the limited resolution of these images further reduced the relationships between macropore porosities and BTC parameters.

Conclusions

Soil macropore characteristics and parameters were quantified using a medical CT and Fiji software, and their effects on preferential flow were tested with a nitrate-leaching experiment. Increased levels of human disturbance (vegetation destruction, cultivation and trampling) with a decrease in altitude resulted in lower macroporosity, surface area density, mean macropore size, macropore number density, length density and node density, but did not result in differences in the mean angle and mean tortuosity of macropores. The macropore distribution was mainly affected by the RLD, soil BD, soil texture and SOM content. Vegetation destruction reduced the RLD and SOM content, resulting in poor macropore networks. Soil compaction in the TG also increased soil BD and decreased macroporosity. The high sand content of the LF due to soil erosion resulted in only a few macropores throughout the soil columns.

Reduced human disturbance corresponded to stronger evidence of preferential flow, as indicated by the fast nitrate breakthrough, long tails and asymmetry of BTCs. The BTCs were well fitted with the equilibrium CDE and MIM models for most soil columns of the HF and MF, but the MIM model outperformed the CDE model in fitting BTCs. In contrast, the equilibrium CDE model was more suitable for fitting the BTCs of the TG and LF. Increased macroporosity in the HF, MF and TG corresponded to higher v , v_m , D and D_m values compared with the LF, whereas few differences in λ (λ_{eff}) were identified among the four sites. This study indicated that human disturbance should be minimised to maintain macropore structure and decrease the risk of natural disasters such as flooding, soil erosion and non-point source pollution to the Yangtze River.

Conflicts of interest

The authors declare no conflicts of interest.

Acknowledgements

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